MILITARY SPECIFICATION

CLOTH, FLANNEL, WOOL, UNDERCOLLAR CLOTH

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers one type of a wool flannel cloth.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

PPP-P-1132 - Packaging and Packing of Woolen, Worsted and Wool Blend (Synthetic Fiber; Cotton) Fabrics

MILITARY

MIL-C-43665 - Cloth Wool: Mothproofing Treatment Of

STANDARDS

FEDERAL

FED-STD-4 - Glossary of Fabric Imperfections
FED-STD-191 - Textile Test Methods

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

FSC 8305

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2.2 Other publications.- The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply:

U.S. Department of Agriculture

Methods of Test for Grade of Wool

(Application for copies should be addressed to the U.S. Department of Agriculture, Agricultural Marketing Service, Washington, D. C. 20402.)

Rules and Regulations Under The Wool Products Labeling Act of 1939

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D. C. 20402.)

3. REQUIREMENTS

3.1 Standard sample.- The finished cloth shall match the standard sample for shade and shall be equal to or better than the standard sample with respect to all characteristics for which the standard sample is referenced (see 6.3).

3.2 Materials.-

3.2.1 Wool.- The wool shall be fleece or pulled wool or a combination thereof not lower in grade than 56's U.S. Standard. The use of reprocessed wool and noils of the same grade is permitted.

3.2.2 Yarn.- The wool shall be carded and spun on the woolen system into single yarns for both warp and filling (see 4.4).

3.3 Color.- The color of the finished cloth shall be as specified (see 6.2) and shall match the standard sample. Piece dyeing is permitted (see 6.4).

* 3.3.1 Mothproofing.- The wool stock, if stock dyeing is used, or the cloth, if piece dyeing is used, shall be mothproofed in accordance with MIL-C-43665.

3.3.2 Matching.- The color of the finished cloth shall match the standard sample under natural (north sky) daylight or artificial daylight having a color temperature of 7500° Kelvin and shall be a good approximation to the standard sample under incandescent lamplight at 2800° Kelvin.

3.3.3 Colorfastness.- The finished cloth shall show fastness to wet dry-cleaning, perspiration, and crocking equal to or better than the standard sample. When no standard sample has been established or designated as applicable to colorfastness, the finished cloth shall show "good" fastness to wet dry-cleaning and perspiration and shall show a Munsell Value for crocking not lower than 8.5. Tests shall be as specified in 4.4.

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3.4 Physical requirements.- The finished cloth shall conform to the requirements specified in table I when tested as specified in 4.4.

TABLE I.- Physical requirements

Weight in ounces (based on linear yard of 56-inch cloth)		Yarns per inch (minimum)		Breaking strength (minimum) (pounds	(pounds)	Weave
Minimum	Maximum	Warp	Filling	Warp	Filling	
10.5	12.5	26	23	24	16	Plain

3.4.1 Width.- The width of the finished cloth shall be a minimum of 60 inches exclusive of selvage (see 4.2.2.2).

3.5 Finish.- The cloth shall be scoured, fulled, (carbonized if necessary) and have a face evenly sheared. The finished cloth shall be pressed and shall have a finish equal to that shown by the standard sample.

3.5.1 pH.- The pH value of the water extract of the finished cloth shall be not lower than 4.0 nor higher than 8.0 when tested as specified in 4.4.

3.6 Length and put-up.- Unless otherwise specified (see 6.2) the cloth shall be furnished in continuous lengths, each not less than 50 yards. The cloth shall be put up in full width rolls as specified in PPP-P-1132.

3.7 Wool content.- The finished cloth shall contain not less than 95 percent wool based on the dry weight of the specimen when tested as specified in 4.4.

3.8 Wool content label.- Cloth manufactured under this specification shall be labeled in accordance with the Wool Products Labeling Act of 1939 (see 2.2).

3.9 Marking.- The back of the cloth shall be marked "Back" at both ends of each roll with letters not less than 1/2 inch in height (see 4.2.2.2). The marking shall be clearly legible in any indelible marking medium used commercially.

3.10 Workmanship.- The finished cloth shall conform to the quality and grade of product established by this specification. The occurence of defects shall not exceed the applicable acceptable quality levels.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in

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the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Inspection.- Sampling for inspection shall be performed in accordance with MIL-STD-105, except when otherwise indicated hereinafter.

4.2.1 Component and material inspection.- In accordance with 4.1 above, components and materials shall be tested in accordance with all the requirements of referenced specifications, drawings and standards unless otherwise excluded, amended, modified or qualified in the specification or applicable purchase documents.

4.2.1.1 Wool.- Examination for grade (see 3.2.1) shall be performed visually by comparison with the applicable U. S. Standard. The wool components shall be examined prior to blending. In the event of dispute resulting from comparison, the grade shall be determined by a width method (wedge).

4.2.1.1.1 Fleece or pulled wool.- Fleece or pulled wool shall be examined in scoured form prior to dyeing, picking, or carding. The examination for grade shall be performed on a composite sample of 1 pound of this component for each 10,000 pounds or fraction thereof in the lot. The composite sample shall be selected from not less than 10 percent of the bales or bags in the lot. The lot shall be unacceptable if the sample does not conform to the requirements of 3.2.1.

4.2.1.1.2 Noils.- Noils shall be examined prior to blending. Sampling and sample size shall be as specified in 4.2.1.1.1.

4.2.1.1.3 Reprocessed wool.- Reprocessed wool shall be examined for grade when reduced to a fibrous state, prior to blending with other component fibers. Method of sampling and sample size shall be as specified in 4.2.1.1.1.

4.2.1.2 In-process inspection.- Inspection shall be made at any point during any phase of the manufacturing process to assure that only approved wools are used (see 3.2.1). The Government reserves the right to exclude from consideration for acceptance any material for which in-process inspection has indicated nonconformance.

4.2.2 Examination of the end item.- Defects found during this examination shall be classified in accordance with 4.2.2.1 through 4.2.2.3.2.

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* 4.2.2.1 Yard-by-yard examination.- The required yardage of each roll shall be examined and defects clearly noticeable at normal inspection distance (3 feet) shall be classified as listed. The defects found shall be counted regardless of their proximity to each other, except when two or more defects represent a single local condition of the cloth, only one shall be scored. A continuous defect shall be counted as one defect for each yard or fraction thereof in which it occurs. The sample unit shall be 1 yard, full width of the finished cloth. The acceptable quality level shall be 4.0 defects per 100 units (yards). The sample size shall be in accordance with inspection level I of MIL-STD-105. The lot size shall be expressed in units of one yard each. The number of rolls selected shall be in accordance with table II.

Defects
Hole, cut or tear
Bad mend, weak place
Smash, multiple floats
Two or more adjacent missing or broken yarns
Knot 1/
Grease or oil stain
Embedded crease or wrinkle

1/ To be scored when larger than limit shown for knots on figure 4 of FED-STD-4.

* 4.2.2.2 Overall examination.- Each defect shall be counted no more than once in each roll examined. The sample unit shall be one roll. The sample size (number of rolls selected as sample) and the acceptance number shall be as shown in table II.

Defects
Width less than specified 1/
Overall uncleanness 1/
Objectionable odor
Degree and character of finish not equal to standard sample
Offshade, shaded side to side, side to center or end to end
Back stamping missing from either or both ends
Back stamping on wrong side
Letters less than 1/2 inch in height
Not labeled or ticketed in accordance with the Rules and Regulations Under the Wool Products Labeling Act

1/ Shall not be scored unless it occurs for more than 5 yards per roll.

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TABLE II.- Sample size

Lot size (yards)	Sample size (rolls)	Acceptance number
Up to 3200 1/	3	0
3201 up to and including 10,000	5	0
10,001 up to and including 35,000	8	0
35,001 up to and including 150,000	13	1
150,001 and over	20	2

1/ If a lot contains fewer than three rolls, each roll in the lot shall be examined.

4.2.2.3 Length examination.-

* 4.2.2.3.1 Individual rolls.- The rolls shall be examined for gross length. Any gross length found to be less than the minimum specified or more than 2 yards less than the gross length marked on the ticket shall be considered a defect with respect to length. The sample unit shall be one roll. The sample size (number of rolls selected as sample) and the acceptance number shall be as shown in table II.

4.2.2.3.2 Total yardage in sample.- The lot shall be unacceptable if the total of the actual gross lengths of rolls in the sample is less than the total of the gross lengths marked on the roll tickets. The rolls examined shall be those selected for the examination of individual rolls.

4.3 Examination of preparation for delivery requirements.- Examination shall be made in accordance with the provisions of PPP-P-1132 to determine that packaging, packing and marking comply with the section 5 requirements.

4.4 Tests.- The methods of testing specified in FED-STD-191, wherever applicable, and as listed in table III shall be followed. The physical and chemical values specified in section 3 apply to the results of the determinations made on a sample unit for test purposes as specified in the applicable test method. The sample unit shall be 1 yard full width of the finished cloth. The lot shall be unacceptable if one or more sample units fail to meet any test requirements specified. The lot size shall be expressed in units of one yard. All test reports shall contain the individual value utilized in expressing the final result. The sample size (number of sample units) shall be in accordance with the following:

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Lot size (yards)	Sample size
800 or less	2
801 up to and including 22,000	3
22,001 and over	5

TABLE III.- Test methods

Characteristics	Requirement paragraph	Test method
Yarn - single		
Warp	3.2.2	Visual 1/
Filling	3.2.2	Visual 1/
Mothproofing	3.3.1	2/
Colorfastness to:		
Crocking	3.3.3	5651
Wet drycleaning	3.3.3	5622
Perspiration	3.3.3	5680
Weave	3.4	Visual 1/
Weight	3.4	5041
Yarns per inch		
Warp	3.4	5050
Filling	3.4	5050
Breaking strength		
Warp	3.4	5100
Filling	3.4	5100
pH	3.5.1	2811
Wool content (finished cloth)	3.7	2101

1/ One determination per sample unit and the result reported as "pass" or "fail".

2/ As specified in MIL-C-43665.

5. PREPARATION FOR DELIVERY

5.1 Put-up and packaging.- Put-up and packing shall be level A or C as specified (see 6.2).

* 5.1.1 Levels A and C.- The cloth shall be put-up and packaged in accordance with the applicable requirements of PPP-P-1132, except that the moth repellent requirement (napthalene flakes) shall not apply.

5.2 Packing.- Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Levels A, B and C.- The cloth shall be packed in accordance with the applicable requirements of PPP-P-1132.

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5.3 Marking.- In addition to any special marking required in the contract or order, shipments shall be marked in accordance with the requirements of PPP-P-1132.

6. NOTES

6.1 Intended use.- The cloth covered by this specification is intended for use as a facing for the underside of the collar of the uniform coats and overcoats.

6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Color of cloth required (see 3.3).
(c) Minimum length if other than specified (see 3.6).
(d) Selection of applicable levels of put-up, packaging, and packing (see 5.1 and 5.2).

6.3 Standard sample.- For access to standard sample, address the procuring office issuing the invitation for bids.

6.4 A suggested but not mandatory dyestuff formulation for shade Army Green 48 is as follows:

Acid Green 58
Acid Green 70
Acid Orange 85
Acid Orange, 86

6.5 Marginal notations.- The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and suppliers are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:

Army - GL
Navy - SA
Air Force - 11

Review activities:

Army - MD
Air Force - 82

User activity:

Navy - MC

Preparing activity:

Army - GL

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